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RESEARCH NOTE

An Upper Carboniferous–Lower Permian (280 Ma) palaeomagnetic pole from the western margin of the Bohemian Massif

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SUMMARY

In the area of the drill site of the German Continental Deep Drilling Project KTB on the western margin of the Bohemian Massif 15 sites of Uppermost Carboniferous to Lowermost Permian rocks (quartz porphyries, granites and a siltstone, geographic mean coordinates: 12°E, 49.8°N) have been studied in order to obtain a reliable pole position for this time interval. The investigated rocks are syngenetic or only slightly younger than the youngest granites of the Fichtelgebirge for which a K/Ar and Rb/Sr age of 280 Ma has been determined by Besang *et al.* (1976). While four sites (quartz porphyries) showed too weak intensities of the natural remanent magnetization NRM, the results of two sites with otherwise good rock magnetic properties had to be discarded because of uncertain tectonic position in a shear zone. The remaining nine sites have all reversed polarity as expected for a remanence age within the reversed Kiaman interval and a mean pole position at 165.9°E, 42.3°N with $N = 9$, $R = 8.8776$, $A_{95} = 6.4°$. This is in good agreement with the mean Upper Carboniferous–Lower Permian European pole position of Stable Europe according to Irving & Irving (1982) and of the Bohemian Massif (Krs 1968). Rock magnetic investigations revealed that the ChRM of the siltstone is due to primary magnetite, while the mostly single-component characteristic remanent magnetization ChRM of the magmatic rocks is carried by primary magnetite and by primary haematite produced by syngenetic high-temperature oxidation of primary magnetite.

Key words: Bohemian Massif, palaeomagnetism, rock magnetism.

1 INTRODUCTION

Earlier palaeomagnetic measurements in the western part of the Bohemian Massif carried out on mostly Devonian rocks from the Frankenwald area (Kim & Soffel 1982; Bachtadse, Heller & Kröner 1983) showed widespread complete or partial remagnetization of Late Palaeozoic age. It has been argued that the entire Variscan mountain belt in Central Europe was affected by remagnetization (e.g. Edel & Coulon 1984) during the latest stage of the Variscan orogeny by an increased heat flow in connection with the intrusion of granites and the extrusion of large bodies of quartz porphyries. In Central Europe quartz porphyries of this age (around 280 Ma) are abundant in the Black Forest, in the Rhenish Shield and in the Thuringian Forest. In the western part of the Bohemian Massif large quartz porphyry bodies are rather rare. They either occur as small intrusive bodies, intercalated between Upper Carboniferous–Lower Permian sediments (mostly sandstones and siltstones) or as dykes associated with granites. In a palaeomagnetic pilot study quartz porphyries and similar rocks including siltstones have been sampled at 15 sites. It was intended to establish a reliable reference direction for the detection (a) of eventual Postpermian rotational movements of the westernmost part of the Bohemian Massif near the Franconian Line with respect to Western Europe and to the central and eastern part of the Bohemian Massif (Krs 1968) and (b) of a Permo-Carboniferous overprint in palaeomagnetic studies of older rocks in the western part of the Bohemian Massif.

2 SAMPLING

Sampling was done with drilling equipment in the field using the magnetic compass for orientation. The geographic coordinates of the sampling sites, the rock types, the stratigraphic ages, the number of drill cores and specimens

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Table 1. Site number, site names, geographic coordinates, rock type (qp: quartzporphyry; pp: piniteporphyry; ss: siltstone; gp: granite-porphyry; cg: crystalgranite), stratigraphic ages (P_1: Lower Permian; C_u: Upper Carboniferous), N: number of cores, n: number of specimens, NRM: site mean NRM intensity in mA m^{-1} (smallest and largest values added in brackets).

No.	Site name	°E	°N	Rocktype	Age	N/n	NRM	
1	Erbendorf 1	12.08	49.84	qp	P_1	15/28	60.4	(6.3-235)
2	Regenstauf	12.14	49.12	pp	C_u	15/24	0.8	(0.19-6.2)
3	Erbendorf 2	12.03	49.85	qp	P_1	15/21	24.3	(14.5-113)
4	Erbendorf 3	12.03	49.85	qp	P_1	10/20	58.7	(20.9-373)
5	Erbendorf 4	12.02	49.84	ss	P_1	19/34	1.5	(0.02-5.6)
6	Lenau 1	11.98	49.92	qp	P_1	19/33	58.2	(33.4-164)
7	Weiden 1	12.22	49.69	qp	C_u-P_1	18/25	0.096	(0.04-0.4)
8	Weiden 2	12.22	49.69	qp	C_u-P_1	18/30	0.111	(0.03-0.54)
9	Weiden 3	12.22	49.69	qp	C_u-P_1	17/30	0.134	(0.07-0.4)
10	Weiden 4	12.22	49.69	qp	C_u-P_1	16/19	0.138	(0.07-0.53)
11	Karlstein 1	12.16	49.16	qp	C_u	11/16	0.186	(0.07-0.57)
12	Marienthal	12.18	49.22	cg	C_u	15/18	0.251	(0.07-5.9)
13	Karlstein 2	12.16	49.16	qp	C_u	14/25	0.195	(0.05-0.65)
14	Göpfersgrün	12.08	50.07	qp	P_1	12/30	0.525	(0.25-28.6)
15	Lenau 2	11.98	49.92	qp	P_1	7/8	134.0	(35.5-600)

per site and the natural remanent magnetization NRM mean intensities are listed in Table 1. Fig. 1 shows a map with the geographical distribution of the sites with respect to the Franconian Line which separates the metamorphic rocks of the Bohemian Massif from the non-metamorphic Permo-Mesozoic sediments in the west. Four sites (site numbers 1, 3–5) are tilted to NW (20°–40°). Two sites (numbers 6 and 15) are situated within a shear zone at the Franconian Line. All other sites are *in situ*. Some are only a few hundreds of metres east of the Franconian Line.

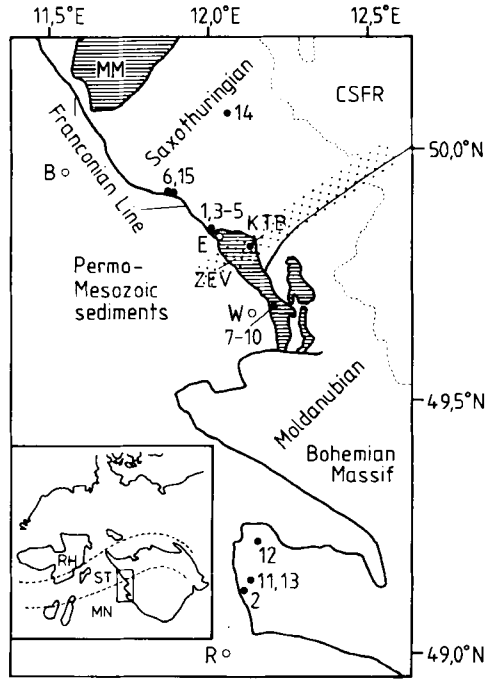


Figure 1. Map showing the sampling sites (closed symbols with site numbers) on the western margin of the Bohemian Massif. R: Regensburg; W: Weiden; E. Erbendorf; B: Bayreuth. Dot: site of the ultradeep drill hole of the German Continental Deep Drilling Project KTB. Curved line and stippled area: supposed contact zone between the Saxothuringian and the Moldanubian. MM, ZEV: nappes of the Münchberger Gneismasse and the Zone Erbendorf–Vohenstrauss, respectively. Insert: position of the investigated area in the Variscan Belt of Central Europe. MN: Moldanubian. ST: Saxothuringian. RH: Rhenohercynian. Dashed line: border between Germany and CSFR.

3 PALAEOMAGNETIC MEASUREMENTS

The NRM of the specimens was measured in part with a Molspin fluxgate spinner magnetometer or with a cryogenic magnetometer of type LETI (Grenoble, France). Fig. 2 shows an equal area projection of the NRM of site 1 (2a) and the site mean NRM directions together with their cones of 95 per cent confidence (2b). In some cases there was

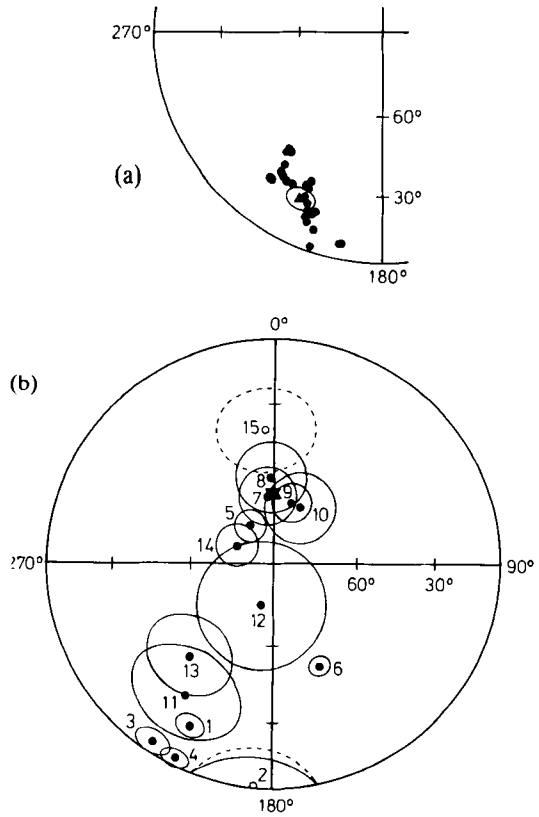


Figure 2. Equal area projection of NRM directions together with the cones of confidence. Numbers refer to site numbers (see Table 1). Closed (open) symbols: positive (negative) inclination. Closed (dashed) ovals of confidence are on the lower (upper) hemisphere. (a) NRM directions of specimens from site number 1. Triangle: mean direction together with oval of 95 per cent confidence. Note the within-site great circle distribution. (b) Site mean NRM directions. Note the between-site great circle distribution.

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already excellent grouping of the NRM far away from the present field direction in the sampling area ($D = 0°$, $I = 65°$; star in Fig. 2b) around directions with $D = 180°$ and mostly shallow inclinations indicating the presence of a significant characteristic remanent magnetization ChRM component. Similar behaviour was found in the Bolzano and other quartz porphyries (e.g. Zijderveld & Van der Voo 1973) and seems to be typical for this type of rock. Sites 7–10 had very weak NRM intensities with directions close to the present-day geomagnetic field direction (Fig. 2b) which collapsed quickly during thermal demagnetization. The NRM mean directions are distributed along a great circle connecting the present geomagnetic field direction with the ChRM direction (Fig. 2b).

Alternating field demagnetization was not successful in the complete destruction of the NRM. Thermal demagnetization was preferred for all types of rocks using a non-magnetic SCHONSTEDT furnace. Fig. 3 shows an example (Zijderveld diagram) of an orthogonal vector plot during thermal demagnetization. In most cases straight lines (like for the specimen from site 4) towards the origin could be obtained after the removal of a component with low blocking temperatures. Complete demagnetization was obtained in the quartz porphyries and the other magmatic rocks only at temperatures close to the Curie temperature of haematite. NRM directions did not change when the Curie temperature of magnetite was passed at 580 °C indicating that magnetite as well as haematite carry the same remanence component (see also Section 4). Site 5 (siltstone) showed blocking temperatures which did not exceed the Curie temperature of magnetite. The NRM components with very low blocking temperatures below 200 °C of possibly viscous origin are oriented more or less in the directions of the present local geomagnetic field. The determination of the single-component ChRM direction was therefore quite simple and did not require a sophisticated multicomponent analysis. It was sufficient to fit the straight section (including the origin in most cases) of the Zijderveld diagram over an extended range of blocking temperatures. The ranges are listed in Table 2. The bedding corrected ChRM data are also listed in Table 2 together with the statistical parameters, the site mean ChRM direction and pole position and the mean pole position (including statistical parameters) for nine sites. A significant fold test could not be made because only four sites are tilted (20°–40° NW) and all other sites are *in situ*. Sites 6 and 15 were discarded because of unreliable tectonic position.

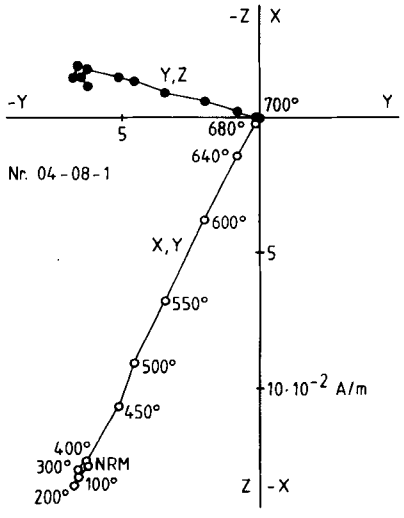


Figure 3. Zijderveld diagram for the thermal demagnetization of a specimen from site 4 (quartz porphyry) before core correction. Open (closed) symbols: projection into the horizontal (vertical) plane. Temperatures are in °C.

Table 2. (a) Directions of bedding corrected characteristic remanent magnetization (ChRM) together with the range of blocking temperature used for the determination of the ChRM and the statistical parameters (n: number of specimens used for the computation of the mean; k, K: precision parameter; α_{95}, A_{95}: semi-opening angle of the 95 per cent cone of confidence). For sites 7–10 no consistent ChRM component could be determined. For the calculation of the ChRM mean direction sites 6 and 15 were discarded because of uncertain tectonic position. (b) Site mean palaeomagnetic pole positions and resulting regional mean pole position. The principal carriers of remanence are also given. pH: primary haematite; pM: primary magnetite; G: goethite; TM: titanomagnetite, Il: ilmenite.

(a)

No.	D(ChRM) (°E)	I(ChRM) (°)	TH (°C)	n	k	α_{95}(°)
1	197.1	+ 8.5	200-680	23	107.0	2.9
2	204.8	- 23.1	200-680	18	30.5	6.4
3	206.6	- 2.1	250-680	16	250.0	2.3
4	202.6	- 6.5	200-680	11	355.0	2.4
5	192.7	+ 13.6	200-580	15	48.1	5.6
6	162.1	+ 34.9	300-680	26	82.1	3.1
11	203.2	- 26.7	300-680	9	65.3	6.4
12	196.5	- 16.9	300-680	9	53.2	7.1
13	203.2	- 18.2	200-680	13	106.0	4.0
14	185.6	- 9.8	350-680	13	207.0	2.9
15	84.0	+ 41.8	400-680	6	43.6	10.3

Mean ChRM direction: D = 199.0° ; I = -9.1° ;
N = 9 ; R = 8.7142 ; k = 28.0 ; α_{95} = 9.9° .

(b)

No.	VGP-LONG. (°E)	VGP-LAT. (°N)	Ores
1	171.4	33.9	pH, TM, Il
2	154.7	47.6	pH, TM, Il
3	158.3	36.2	pH, TM, Il
4	162.2	39.6	pH, TM, Il
5	177.1	32.2	pM
11	155.6	50.1	pH
12	167.8	47.1	pH
13	158.3	45.7	pH
14	184.2	44.6	pM, G

Mean (north)pole position:
= 165.9°E ; = 42.3°N ;
N = 9 ; R = 8.8776 ;
K = 65.4 ; A_{95} = 6.4° .

4 ROCK MAGNETIC MEASUREMENTS AND ORE MICROSCOPY

The acquisition of saturation remanence (SIRM-curves) has been determined for specimens from all sites in order to obtain information about the main ferromagnetic ores (Dunlop 1972). Three types of SIRM-curves have been determined for which examples are shown in Fig. 4. The first type was observed for the quartz porphyries (Fig. 4a, specimen from site 4). The curve shows a rapid increase in low fields between 0 and 0.1 T followed by a more gentle convex increase between 0.1 and 1.5 T. We interpret this as

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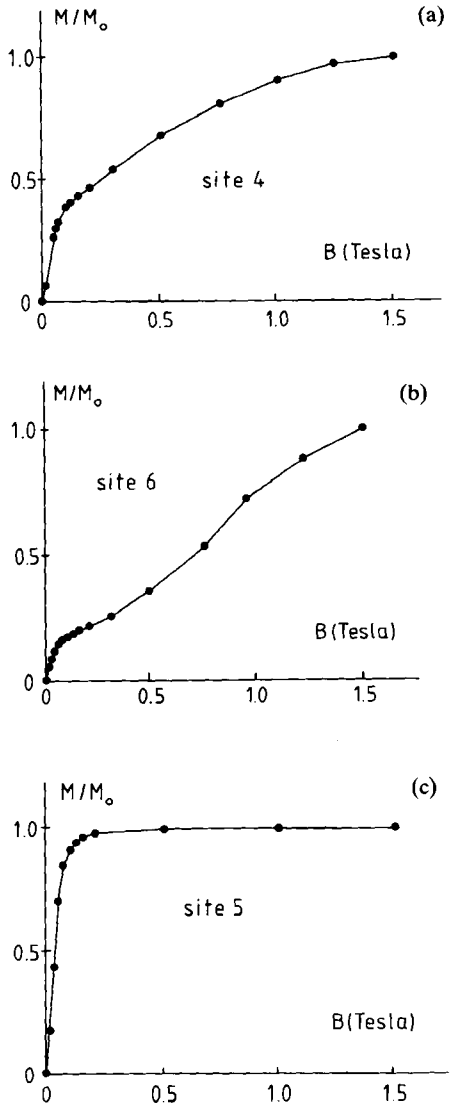


Figure 4. Saturation remanence (SIRM) acquisition curves, normalized to the final intensity obtained in a field of 1.5 T. (a) Specimen from site 4 (quartz porphyry), indication for magnetite and haematite. (b) Altered specimen from site 6 (quartz porphyry situated within a shear zone), indication for magnetite and goethite. (c) Specimen from site 5 (silt stone), indication for exclusively magnetite.

an indication for magnetite (with lower coercivity) and haematite (with higher coercivity). A combination of magnetite (low coercivity) and goethite (extremely high coercivity) is shown in Fig. 4(b) (site 6). The third type of curve was obtained for the siltstone of site 5 (Fig. 4c) which shows the low coercivity typical for magnetite. The specimens for which the SIRM-curves have been determined were also demagnetized thermally. The blocking temperature spectra confirm the presence of magnetite plus haematite for site 4, magnetite and goethite for site 6 and just magnetite for site 5. Curie temperature measurements support these results as well.

The nature of the ferromagnetic mineral assemblages was also investigated by ore microscopy. Of special interest was the question whether the haematite was of secondary (e.g. due to alteration) or of primary origin. Fig. 5 shows a polished section with a typical ore grain from site 1 under crossed Nicols. The rather large ore grain consists of grey lamellae of magnetite in a darker grey ilmenite ground mass. This is typical for a high-temperature oxidation and exsolution process. The haematite (light grey) has in part replaced the magnetite syngenetically by an oxidation process which affected also the silicates leading to a red colour of the quartz porphyries. They resemble in this respect and also in respect to their rockmagnetic and palaeomagnetic properties the Bolzano quartz porphyries (Zijderveld & Van der Voo 1973). The thermal demagnetization experiments (Fig. 3) in connection with the optical examination under reflected light indicate that the ChRM carried by exsolved magnetite and by haematite is syngenetic. However, in some of the material the magnetic minerals were too small to be identified by standard microscopic means. In some cases minute amounts of titanomagnetite and ilmenite could also be detected. However, the abundance of titanomagnetite compared to haematite is so low that it does not contribute significantly to the remanence of the rocks. Some goethite was only found in altered samples like from site 6 (Fig. 4b) and occasionally from site 14. In the siltstones, on the other hand, only very fine-grained magnetite and no other ferromagnetic mineral could be detected, which is in agreement with the conclusions from the SIRM acquisition curve of Fig. 4(c). The optically observed main ferromagnetic ores are also listed in Table 2. In conclusion we assume that the ChRM of the porphyries is carried by both primary magnetite and haematite, while that of the siltstone is due to primary magnetite alone.

5 PALAEOMAGNETIC POLE POSITION AND CONCLUSIONS

The mean pole position of the nine sites (Table 2) is situated at 165.9°E, 42.3°N, $A_{95} = 6.4°$ and is plotted in Fig. 6 in equal area projection together with the apparent polar wander path (APWP) of Stable Europe according to Irving & Irving (1982) for the time interval 230–330 Ma. The result of this paper (star) matches the Upper Carboniferous–Lower Permian APWP and coincides with the mean 270 Ma pole position (not plotted in Fig. 6). It is also close to the pole (square) of the secondary (overprint) remanence direction (164.4°E, 45.7°N, $A_{95} = 5.5°$) determined by Kim & Soffel (1982) for the Devonian rocks in the Frankenwald area about 60 km north of Erbendorf. There is also agreement with the Upper Carboniferous–Lower Permian data compiled by Krs (1968) for the central and eastern part of the Bohemian Massif. Several conclusions can be drawn from this study.

(a) It gives a reliable reference direction for the Permo-Carboniferous in the western part of the Bohemian Massif, which is useful for a future palaeomagnetic study of Lower Palaeozoic and older rocks in the area.

(b) The above-mentioned radiometric age determinations carried out on the latest granite bodies in the Fichtelgebirge (Besang *et al.* 1976) point to a maximum age of the investigated rocks of about 280 Ma. According to G. Stettner (1990, private communication) the age of the quartz

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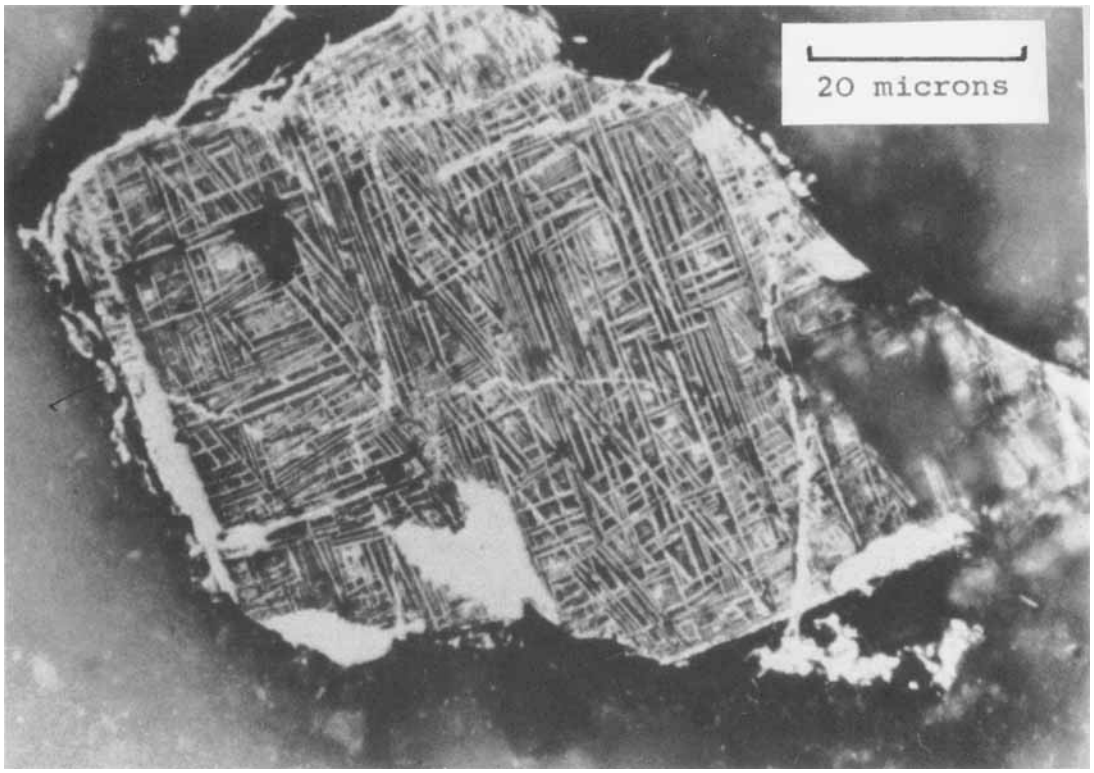


Figure 5. Polished section of a large ore grain from site number 1 under crossed Nicols with exsolution lamaellae of magnetite (grey) in an ilmenite magnetite ground mass (dark grey) and haematite (light grey). Scale bar: 20 μm.

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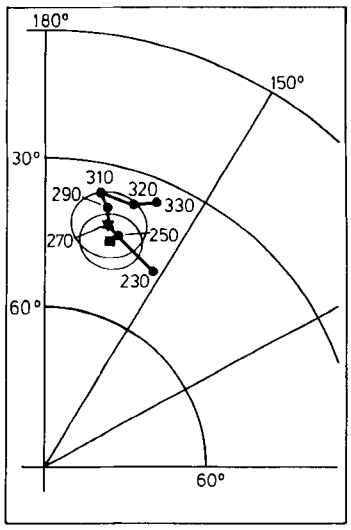


Figure 6. European apparent polar wander path for the time interval 230–330 Ma (after Irving & Irving 1982) in equal area projection with the mean pole positions (together with the A_{95} cones of confidence) of this study (star) covering the 270 Ma data point. Square: result of the study by Kim & Soffel (1982) on Devonian rocks from the Frankenwald area north of the Münchberger Gneismasse (MM in Fig. 1) showing a Permo-Carboniferous overprint.

porphyries in the western part of the Bohemian Massif is between 275 and 280 Ma (Lowermost Autunian). This age is in agreement with the remanence age concluded from the APWP. Radiometric age determinations on the quartz porphyries are planned.

(c) The palaeomagnetic result indicates that there was no relative rotation of the western margin of the Bohemian Massif with respect to its central and eastern part confirming the hypothesis of Krs (1968) that this unit behaved as a rigid block during the last 280 Ma and that there were no rotational movements of the Bohemian Massif with respect to Stable Europe since that time.

(d) The uplift of the Bohemian Massif since the Cretaceous did not affect the western margin of the Bohemian Massif in terms of local rotations or other deformations even in places very close to the Franconian Line. Only two sites (6 and 15) situated in a shear zone within the Franconian Line (Fig. 1) show indications of a left lateral movement and some tilting.

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